



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

STATISTICS

PAPER 1

4073/1

2 hours 30 minutes

SPECIMEN PAPER

Additional materials:

Answer booklet

Mathematical Instruments

Graph paper

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces at the top of each page.
Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.

Answer **all** questions.

All working must be clearly shown.

Omission of essential working will result in loss of marks.

Decimal answers which are not exact should be given to three significant figures unless stated otherwise.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
This paper is marked out of **100**.

This specimen paper consists of 6 printed pages.

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- 1 (a) State the **two** branches of Statistics. [2]

- (b) Below are the ages of 20 learners at a junior school.

9 5 7 7 5 12 6 6 12 5 7 7 5 5 4 7 4 7 9 6

Copy and complete the table.

Age	Tally	Frequency
4		-
5	-	5
6	-	-
7		6
9		-
12	-	-

[3]

- 2 (a) Give any **one** example of the following types of data:

- (i) Discrete data [1]

- (ii) Continuous data [1]

- (b) The ages (x) of students in a secondary school choir lies between 14 and 20.

- (i) State the lower limit of the ages. [1]

- (ii) Write the age limits in the form $a \leq x < b$, where a and b are constants. [2]

- 3 (a) Write down any **one** advantage and any **one** disadvantage of using mean as a measure of central tendency. [2]

- (b) The table shows the number of novels that were completed by form 4 students at a certain school.

Number of novels	1	2	3	4	5	6
Frequency	3	9	12	11	8	7

- (i) State the mode. [1]

- (ii) Calculate the mean. [3]

4 Define the following terms as they are used in statistics.

(a) Sample [2]

(b) Population [2]

(c) Randomness [2]

5 The probability distribution of a discrete random variable X is given in the table.

X	1	2	3	4
$P(X=x)$	k	$4k$	$9k$	$16k$

(a) Find the value of k . [2]

(b) Hence calculate the probabilities and complete the table.

x	1	2	3	4
$P(X=x)$				

[2]

(c) Calculate $P(X < 3)$ [2]

6 (a) State any **two** sources of errors. [2]

(b) Tapiwa measured the mass of a chicken from the farm. The mass of the chicken was 2,1 kg instead he got 2,3 kg.

Find the

(i) absolute error, [2]

(ii) relative error. [2]

7 (a) State any **two** measures of dispersion. [2]

(b) The number of tonnes of maize harvested by a farmer in 5 consecutive years are as follows: 11; 14; 17; 23; 29.

Calculate the

(i) range, [2]

(ii) variance. [3]

- 8 (a) Define the term **price relative**. [2]
- (b) The following table shows the price, in \$, of sugar and rice in 2020 and 2024.

	2020	2024
Sugar	2,30	2,50
Rice	1,50	2,00

Taking 2020 as the base year, calculate the price relative for

- (i) sugar, [2]
- (ii) rice. [1]
- (c) Find the simple aggregate cost index to the nearest whole number. [2]
- 9 A woman records the number of chickens and pigs she sells each day at her farm and the information is illustrated in the table.

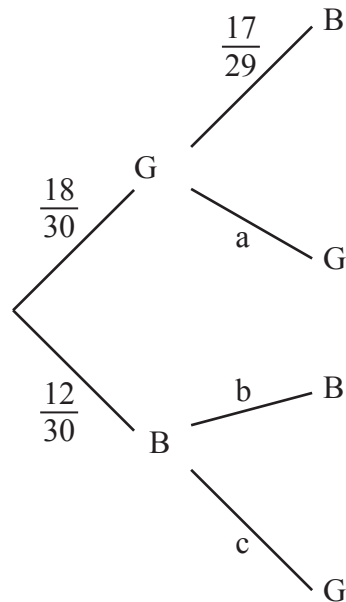
Number of chickens (X)	5	7	12	16	20
Number of pigs (Y)	4	12	18	21	24

- (a) Using a scale of 2 cm to represent 5 units in the horizontal axis, and 2 cm to represent 2 units in the vertical axis, draw a scatter diagram to represent the data. [4]
- (b) Calculate $\bar{x}$ and $\bar{y}$. [3]
- 10 Given that the first 4-point moving averages of the set

17 16 x 14 18 17 19
is 15.5

- (a) find the
- (i) value of x , [3]
- (ii) second 4-point moving averages for the set. [2]
- (b) Calculate the first 3-point moving average for the set. [2]

- 11** Two students are randomly selected from a class to visit the Great Zimbabwe monuments. The class consists of 18 girls (A) and 12 boys (B). Some of the possible outcomes and their probabilities are shown in the tree diagram.



- Find the
- (a) values of a, b and c. [4]
- (b) probability of selecting a boy and a girl to the trip. [3]

- 12** (a) Define the term **probability**. [1]
- (b) A fair tetrahedral (4-sided) die is thrown, and a coin is tossed.

- (i) Copy and complete the outcome table.

	1		2	3	4
Coin	H	H		H 3	
	T	T 1	T 2		T 4

[3]

- (ii) Calculate the probability of obtaining a head and an even number. [3]

- 13** Below are shoe sizes worn by 10 randomly selected workers at a certain company

6, 9, 7, 5, 7, 8, 8, 9, 6, 8

- (a)** Copy and complete the frequency table below for the data.

Size	5	6	7	8	9
Frequency			2		

[2]

- (b)** Draw a pie chart to represent the information in **(a)**.

[6]

- 14** State the **four** components of time series and give one example for each.

[8]

- 15** The information in the following table relates to a certain province.

Age group	Population	Number of deaths	Age specific death rate	Standard population %
0 – 19	35 000	160	4.6	25
20 – 39	60 000	x	2.5	30
40 – 59	y	380	7.6	25
60 and over	60 000	770	12.8	20

- (a)** Calculate the value of x and the value of y .

[4]

- (b)** Giving answers correct to two decimal places, calculate the standardized death rate for the province.

[4]